Oliver Ma

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Education

University of California, Berkeley — Berkeley, CA

Bachelor of Science in Electrical Engineering & Computer Sciences — *Expected Graduation: May 2028*

- **GPA**: 3.91/4.00 (Dean's List - College of Engineering)
- **Relevant Coursework**: Computer Programs, Data Structures, Computer Architecture, Signals, Data Science
- **Honors & Awards**: Citadel Trading Challenge Winner, Jane Street Guts++ Competition 2nd Place

Experience

Anyscale — Sep. 2026 – Present

Forward Deployed Engineer, Contract — *San Francisco, CA*

- Build autonomous vehicle testbed for Anyscale's robotics lab, cutting platform cost by **70%** by programming 1/17 scale car on NVIDIA Jetson Orin Nano to run perception models integrating camera and IMU sensor data
- Reduce onboard power draw by **25%** with **<3% accuracy loss** across **12 model configurations** by building energy-monitoring pipeline logging power and GPU utilization at **50 Hz** across **5 telemetry channels**
- Develop computer vision perception system achieving **88% lane detection accuracy** at **15 FPS** on onboard hardware by training lightweight CNN models on **3000+ labeled frames** and optimizing with quantization

Capital One — June 2026 – Aug. 2026

Software Engineering Intern — *McLean, VA*

- Delivered contract-first Go microservice and REST API serving year-end rewards data to **100M+ Capital One cardholders**, replacing static email with interactive web feature projected to increase login traffic by **∼15%**
- Designed pre-computation ETL pipeline using PySpark, AWS Glue, and DynamoDB to process **10B+ raw data rows** into structured response models, cutting API response time from a projected **∼4 hours** to **66 minutes**
- Containerized Go service with Docker and deployed to AWS ECS Fargate via CloudFormation and internal CI/CD pipeline, enabling zero-downtime builds and reducing release effort from **∼2 hours to under 10 minutes**

Verizon — June 2023 – July 2023

Data Engineer Intern — *San Jose, CA*

- Architected and deployed full-stack DevOps monitoring dashboard integrating Jenkins and Jira REST APIs
- Developed real-time data visualization features enabling administrators to monitor **20+ KPIs** across CI/CD pipelines and project backlogs, reducing workflow time by **∼40%** by consolidating metrics into single interface

Projects

Character-Level Transformer with KV-Cache and Speculative Decoding | *Python, PyTorch, CUDA*

- Implemented a **10.8M-parameter** decoder-only transformer and character-level tokenizer using manual Q/K/V, scaled dot-product attention, causal masking, and pre-LN blocks, without using library transformer blocks
- Trained to **1.44 validation cross-entropy** in **236 seconds** on a **628K-character corpus** parsed from **17K Discord messages** using per-layer KV caching and speculative decoding with a training-free n-gram drafter

OncoVision | *Svelte, SvelteKit, Tailwind CSS, TensorFlow, Supabase*

- Engineered progressive web application achieving **85%+ accuracy** in oral and skin cancer risk detection from camera scans using fine-tuned computer vision models, securing **3rd Place out of 50+ teams** at E-Med Hacks
- Optimized **∼4MB MobileNetV2 model** via post-training quantization for real-time inference at **<150ms per scan**, enabling instant health screening on mid-range Android and iOS hardware without cloud dependencies

Publications

The Comprehensive Guide to Training and Running YOLOv8 Models on Custom Datasets, *Towards Data Science* — 02 Oct. 2024

Technical Skills

Languages: Python, Java, C/C++, Go (Golang), SQL, RISC-V Assembly, JavaScript, TypeScript, HTML/CSS
Frameworks & Libraries: React, Angular, Next.js, SvelteKit, Tailwind CSS, NumPy, PyTorch, TensorFlow, OpenCV
Developer Tools: Git, GitHub, Docker, Jenkins, Jira, Amazon ECS, Amazon EC2, Amazon DynamoDB, AWS Glue